



**Design and Development of a Reliable
Automated Pet Enclosure with Feeder Jam
Detection, Fail-Safe Ventilation System and Pet
Health Monitoring in Sri Lanka**

Department of Information and Communication Technology

Faculty of Technology

South Eastern University of Sri Lanka

Group Number:

07 Network

Group Members:

SEU/IS/20/ICT/014 M.T. Madapatha

SEU/IS/20/ICT/015 M.B. Jayamanne

SEU/IS/20/ICT/018 S.T.P. Edirisinghe

SEU/IS/20/ICT/020 K.P.S.D. Karunarathna

SEU/IS/20/ICT/061 D.R.S.D.S. Wijayasinghe

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1. Introduction

Significantly increasing numbers of pet owners have been reported throughout the country, leading to an increasing demand for effective and safe methods of providing pet care. There are new automated systems available to enable pet owners to keep track of their pets' feeding schedules, environments, and behavior patterns. This will help alleviate some of the physical effort required of pet owners and improve their animals' health and welfare.

Many current livestock production systems are primarily focused on functional automation as opposed to providing for the operational reliability of the system. Research findings indicate that two major problems are regularly observed with feeder malfunction resulting from mechanical jamming and ventilation failure during periods of high ambient temperature. In the case of feeder jamming, food cannot be dispensed from the feeder, and should the ventilation system cease to function for any reason, the animals will not receive any cooling and experience overheating, dehydration, and stress to such an extent that it will result in mortality or extreme distress to the animals. The risk of these failures occurring is high (especially in Sri Lanka and other countries with a tropical climate).

The aim of this study is to develop a reliable, self-sufficient, automated pet enclosure capable of detecting blockages in the feeding apparatus and breakdowns of the ventilation unit. In case of either event, the fail-safe system will react with measures to solve the problem, then trigger the transmission of notification alerts. With this project, we hope to provide pet owners in Sri Lanka with a sustainable and reliable means of securing their pets and reducing environmental risks that can be detrimental to their pets' health as well as the health of their owners and families.

2. Literature Review

Existing literature highlights notable progress in automated pet care and animal monitoring systems, driven by the increasing demand for remote management and improved animal welfare. The use of smart cages and enclosures based on the Internet of Things has been well documented, including studies that examine IoT-based Smart Cages and Enclosures (Nan et al., 2025; Ng et al., 2017; Technology, 2024), and multiple applications that provide significant waste management solutions (Department of Computer Science and Technology, Uva Wellassa

University, Badulla, Sri Lanka et al., 2019; Vedhashree et al., 2025), automatic feed systems (Durga Prasad et al., 2021; Tippannavar et al., 2023), and video surveillance. Furthermore, there have been major advances in feeding systems, with basic feed dispensers operated by smartphone apps to solar-powered units, as well as sophisticated feed systems equipped with deep learning algorithms to enable identification of individual pets and daily portioning of food (Airikala et al., 2021; Castillo-Arceo et al., 2024). In addition, much has been done with integrated care ecosystems (Dissanayaka et al., 2019; Luayon et al., 2019; Mandy et al., 2016; Rey & Rey, 2024) which control all aspects of caring for pets, such as automatic water dispensing, cleaning of litter boxes, and controlling the opening and closing of doors, typically through a central mobile application for user convenience (L et al., 2020). Finally, similar to smart pet technology, there has been significant success in the development of IoT-based technology for livestock and poultry (Kashmira et al., 2021; Rahmatulloh et al., 2022), utilizing sensors for cleaning manure, collecting eggs (Butt et al., 2023), and monitoring health through the use of remote monitoring systems (Arshad et al., 2022). RFID (Francis & Shah, 2022) and GPS-based LoRa WAN systems (Tharmaseelan et al., 2021) are two examples of technological developments that have been used to improve animal physical security and minimize animal losses (Krueger et al., 2019). However, there are still many constraints associated with existing automated enclosures. Many of these automated confinements are limited in automation and require manual intervention to maintain them. Safety agents that are installed within most automatons can stop the machinery if there is some type of abnormal motion or electro-mechanical current peak; however, they are not incorporated into all (i.e. found in some automated systems). Additionally, conventional environmental control systems are usually designed with simple threshold-based triggers for temperature and humidity with no predictive capability or redundancy thereby presenting a risk to the health of the animals should the system fail (ANALYSIS OF VENTILATION AND THERMAL REQUIREMENTS OF LABORATORY ANIMALS CAGES IN ENVIRONMENTALLY CONTROLLED SPACES-*ProQuest*, n.d.; Chen & Elshakankiri, 2020). Although vision-based systems for disease detection and growth monitoring have shown high accuracy in aquaculture and hydroponics, the application of such robust visual analytics (Jayasinghe et al., 2023) to domestic pet safety remains underutilized in commercial smart cages. Most existing feeders assume successful delivery without mechanisms to confirm output, leaving a gap in reliability for critical care scenarios.

Finally, the deployment of such systems within the context of Sri Lanka presents unique challenges (Lanerolle et al., 2018) and opportunities. The country has seen a surge in IoT

adoption for agricultural efficiency, evidenced by smart systems for paddy cultivation, greenhouse automation (Laksiri et al., 2019; Mallikarathne et al., 2024), and crop recommendation based on soil fertility (Fernando & Gunasekara, 2020). Researchers have successfully applied sensor networks to local problems ranging from elephant intrusion detection and aquatic safety (Pinnagodage et al., 2024) to beehive monitoring (Liyanage et al., 2024) and rubber tapping automation (Yatawara et al., 2019). The existing problems in local infrastructures include a major issue with utilizing real-time environmental data (Sudantha et al., 2020) to create a cost-effective and energy harvesting solution due to the fiscal constraints of the economy. There are existing smart solutions outside of the pet care sector, such as traffic management (Abeynayake & Menoli kaushalya, 2022; De Zoysa et al., 2007), waste management and electricity conservation systems at universities (Vasanthapriyan & Randima, 2019), but the pet care sector relies on traditional manual recording methods or basic monitoring to provide support to domestic animals living in Sri Lanka. As a result, there is a need to change the sensor-based and data-supported systems employed in agricultural production, to be resilient and provide for domestic agricultural animal care (Bandara, n.d.; Samaranyake et al., 2022). that can be managed in a tropical climate without the need for constant human monitoring (Ekanayake et al., 2022; Gunathilaka et al., 2019).

3. Research Gap

Based on the reviewed studies, the following specific gaps are identified and serve as the focus of this research:

3.1. Automatic feeder jamming is insufficiently addressed.

Current systems rarely detect mechanical blockages or verify whether food has actually been dispensed, leading to undetected feeding failure.

3.2. Ventilation failure during heat is inadequately managed.

The majority of systems have neither a backup ventilation system nor an alert system for when the fans fail or when the heat rises beyond the safe limit of operation

3.3. Continuous animal health and activity monitoring is poorly Integrated.

Currently there are wearable systems that exist today. However, a lot of wearables are not able to be powered by renewable energy sources and most do not have real time monitoring capabilities. The Solar-Powered Smart Health Belt addresses these two areas. This one utilizing a reliability-centered design methodology will rectify these deficiencies.

4. Methodology

The proposed system will be developed using a modular architecture consisting of a feeding unit, environmental monitoring unit, control module, and communication interface.

4.1. Feeder reliability mechanism:

The feeding bowl is equipped with a load cell to determine if food has been dispensed, and motor torque and/or current can be measured to identify an obstruction. The system will start an automatic clearance cycle once an obstruction is detected. Should this cycle not solve the problem, the user will receive a notification for assistance, and an alternate method will be put into place for dispensing the item.

4.2. Ventilation reliability mechanism:

Monitoring internal cage temperature and humidity through sensors will be ongoing. Electrical current used by, or fan rotational speed will be monitored to identify any failure in the fans' mechanical operation/failure. If an overheating condition exists, or should the fan stop functioning, then another fan or passive ventilation method will be initiated automatically and a user notification will occur.

4.3. Power and connectivity considerations:

During power outages, battery backup allows for the uninterrupted operation of the device, while essential safety functions remain operational during an Internet disconnection..

4.4. Evaluation

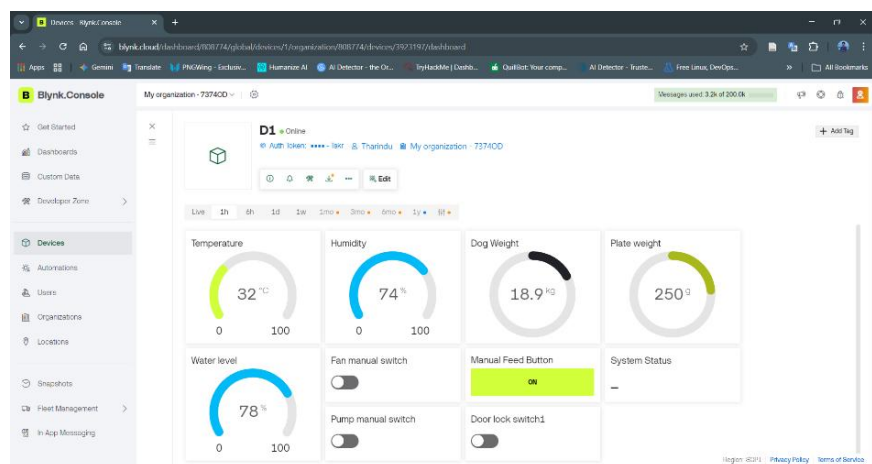
Through experimentation, we will evaluate the effectiveness of jam detection accuracy, ventilation speed, recovery from a failure, and overall stability of the system during an environmental simulation.

4.5. Animal health Activity Monitoring Mechanism

This lightweight (less than 1oz), user-friendly belt allows all dog owners to engage in their everyday routine while remaining active. With built-in Motion Sensors (MS) and Heart Rate Sensors (HRS), this piece of wearable technology gives realtime activity and health information to all owners, as it tracks all movements and gives an indication of the health and activity levels of their dogs. This device can also generate notifications based on the way the dog is moving; identifying possible signs of illness, distress, stress or other abnormal actions. The belt communicates with the ESP32 via wireless transmission, and any notifications generated during abnormal situations will be sent immediately to the owner's smartphone via the ESP32. Powered by a lithium-ion rechargeable battery, able to last several days of normal use and provide solar charging even in low-light conditions as well.

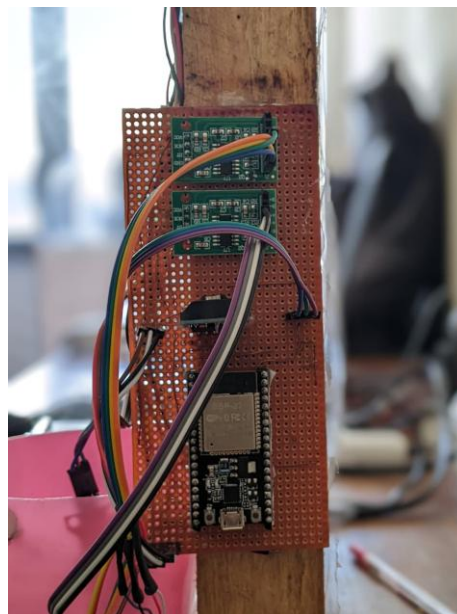
5. Results

Testing was done on the developed Automated Pet Enclosure System in a realistic setting to determine how well each individual component integrated with one another. During this testing phase, all hardware components on the cage (cage modules) and the wireless collar with wearables for monitoring animal health were evaluated. The overall results indicate that the entire system has functioned as desired across all major functional areas.



5.1. Cage Components

The performance of all components installed within the cage was satisfactory and retained reliability through all tests utilized. The response of two ventilation fans to the temperature and humidity readings from the DHT11 Temperature and Humidity Sensor produced accurate activation/deactivation of the devices according to specified thresholds with zero observed failures. The automatic feeder successfully dispensed food at preset intervals, while also detecting the presence of blockages within the feeder's pathway according to its design; in addition, the blockages triggered the clearance sequence, without any need for human intervention. The servo actuated door lock operated consistently as intended for both scheduled commands as well as user-triggered, while communicating with the ESP32 microcontroller without noticeable interruption, including the consistent streaming of real-time values via Wi-Fi from the ESP32 to the Blynk app. Along with streaming data in real-time to the Blynk Dashboard, the dashboard displayed temperature, humidity, feeder status, fan status, and door lock state, allowing for simulated remote monitoring to occur in an effective manner.



5.2. Animal Health Collar Components

The testing of the wearable health monitoring collar with the integrated ESP32 C3 Mini showed positive results. A heartbeat sensor was able to measure pulse of test subjects and a motion sensor utilized an MPU9250 device to gather data on movement and activity patterns in real-time for the test subjects being monitored. Pulse data and activity levels were sent via wireless transmission to the primary ESP32 device where this information was sent to the Blynk app for users to observe pulse and activity level data on the Blynk app's dashboard. A lithium-ion rechargeable battery contained sufficient power for the duration of the testing sessions. The solar charging module was operational based on the available light during testing. The system was able to generate the proper alerts for users during simulated events of abnormal activity levels.

5.3. Limitations

One of the issues identified while testing included performance limits within the water level sensor. Although it was an operational and capable unit that could identify and indicate the presence of water, the actual values reported by the sensor did not have the degree of accuracy needed to conduct threshold-based monitoring at the identified water level. The output values produced by the water level sensor were also inconsistent based on different conditions related to the intermediate levels of the water detected. As a result, the overall reliability of the system for monitoring the presence of water and triggering the refill operation were not as stable as other components within the system.

6. Discussion

By integrating feeder jam detection with fail-safe ventilation and real-time health monitoring into one platform, the project closed the three main research gaps. In Sri Lanka's tropical climate, the dual-fan ventilation worked correctly when triggered by the sensors. The feeder jam detection showed improved functionality over regular feeders that do not report whether or not the feeder has delivered the bird food. The health monitoring collar provided real-time streaming of heart rate and motion to Blynk using solar recharging allowing for long-term viability of the collar. Blynk was proven to work as a centralized control and monitoring

platform. The only major downside of the system was the water level sensor. Due to a low-cost resistive design, it gave inconsistent readings at mid-decimal values, however the problems can be rectified by changing to an ultrasonic or capacitive sensor or by utilizing firmware-side signal filtering. This makes the system a viable, cost-efficient IoT solution and provides aspirations for future development in sensor accuracy, wireless range, and classifier models based on machine learning.

7. Tools and Technologies

- **Microcontroller platform:** ESP32 Development Board
- **Sensors:** HX711 for load cells, DHT11 temperature and humidity, heartbeat sensor, motion sensor (MPU9250), water level sensor
- **Actuators:** servo-driven feeder mechanism and door lock, 2 reliable fans, 2way relay module, Water pump,
- **Communication:** Wi-Fi.
- **Power components:** Battery backup and power regulation circuitry, Mini solar cell, Rechargeable Battery for belt, Power Management Module
- **Software:** Arduino IDE and Blynk software.

8. Budget

Component	Price (Rs.)
ESP32 development board	1490.00
ESP32 C3 mini development board	790.00
Dot board	280.00
Jumper wires	250.00
Load cells (5)	1000.00
HX711 for load cells	260.00

DHT11	340.00
RTC module	150.00
Servo motor (3)	975.00
2way relay	250.00
PC case fan (20)	1580.00
9V battery	200.00
Heart beat sensor	250.00
MPU9250 gyroscope and motion sensor	680.00
5v mini solar panel	350.00
Lithium rechargeable battery	440.00
TP4056 power module	50.00
Water level sensor	60.00
Water pump	160.00
Total	9,705.00

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